

Dynamic Multi-Model LLM Routing via Evolutionary Code Search

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Abstract

Deploying large language models economically requires routing each query to the cheapest model tier that preserves answer quality. We reframe router design as a program-search problem: routing policies are executable Python functions, evolved by an island-model genetic search over a quality-diversity (MAP-Elites) archive, and validated by a deterministic multi-stage oracle comprising an AST security gate, boundary smoke tests, and a vectorized benchmark over 6,000 synthetic queries with per-tier quality and token-price models. Without any LLM-in-the-loop calls, the search discovers a 3-island + migration policy that achieves **81.4% cost reduction** relative to always-frontier routing while retaining 82.0% of frontier quality, at a mean decision latency of **0.138 μ s** per query – a $489\times$ inference-speedup over a supervised decision-tree router of comparable quality. The champion policy generalizes positively under a distribution shift (85.5% cost reduction on OOD queries). The study is framed as a formal proof of concept for the search methodology and its zero-latency AST-evaluated oracle, and the ablation suite is replicated over five seeds: mean peak fitness is remarkably robust to topology, archive, and operator ablations (all variant means within 0.17 fitness, $\sigma \leq 0.43$), with semantic mutation the sole convergence-axis signal (Gen $\rightarrow$ 95% of 1.0 ± 0.0 vs 1.6 ± 1.3 for random jitter) and island topology reducing frontier-size variance by 65%. We release the full lineage database, ablation harness, and paper-generation pipeline.

1 Introduction and Related Work

Commercial LLM APIs span a $20\times$ price gradient between small and frontier tiers, yet query difficulty is highly heterogeneous: a substantial fraction of production traffic can be served by cheap models with negligible quality loss [1]. Learned routers typically frame this as supervised prediction of the strongest-cheapest model [2, 3], which requires preference data, generalizes poorly under shift, and adds an ML inference stage whose latency can rival the routing decision it optimizes. Program-search methods such as FunSearch [4] and Evolution-through-Large-Models [7] instead evolve executable code, inheriting interpretability, verifiability, and near-zero deployment cost.

The latency wall. Routing sits on the critical path of time-to-first-token: whatever the router spends, the user waits. Neural routers embed every query with a transformer encoder before classification, costing tens to hundreds of milliseconds per request on CPU – and non-trivial tail latency even when batched on accelerators – three to five orders of magnitude above a chain of symbolic comparisons. At 10k queries per second, the embedding pass alone becomes a dedicated model-serving cost center. An admissible router must therefore decide in microseconds: any quality recovered by smarter routing is voided if the router itself adds perceptible latency. This constraint motivates evolving *compiled symbolic policies* whose decision cost is a handful of dictionary lookups,

and it is why our oracle hard-penalizes any policy slower than $100\ \mu\text{s}$ – still three orders of magnitude below a single neural embedding forward pass. We treat the router itself as the evolving artifact, and this work as a formal proof of concept: the object of study is the search methodology together with its zero-latency AST-evaluated oracle, deliberately isolated from production concerns so that every claim is exactly reproducible. The deterministic oracle replaces expensive LLM feedback with a simulated but internally consistent query corpus and per-tier quality model, enabling thousands of policy evaluations per minute. Our contributions are: (i) a three-gate evaluation oracle (AST security, smoke tests, vectorized benchmark with hard latency penalty); (ii) an island-model MAP-Elites search over routing-code mutants with full SQLite lineage; (iii) a controlled three-axis ablation (topology, archive, operators), replicated over five seeds; and (iv) a mechanistic deconstruction of the discovered champion policy 22e2a1f5, which never calls the frontier tier yet dominates every hand-designed baseline on the quality–cost frontier.

Related work. FrugalGPT [1] cascades models with escalation; RouteLLM [2] and Hybrid-LLM [3] learn routers from preference pairs; FunSearch [4] evolves programs with LLM mutation operators; MAP-Elites [5] provides the quality-diversity archive we bin over (quality $\times$ cost reduction); NSGA-II [6] defines the non-dominated Pareto bookkeeping used for reporting. Unlike all of the above, our router is a compiled, auditable Python function whose entire decision structure fits in a dozen lines.

2 Problem Formulation and Multi-Stage Deterministic Oracle

Each query q is a feature vector (character length, estimated tokens, code-syntax and math-symbol flags, detected language, readability, semantic density, budget ratio, latency deadline) paired with per-tier qualities $\{\text{qual}(q, m)\}_{m \in \mathcal{M}}$, $\mathcal{M} = \{\text{small, medium, frontier}\}$. A policy $\pi : q \mapsto m$ is a pure Python function `select_model(q)`. Token prices $p^{\text{in}}, p^{\text{out}}$ per tier are (0.15, 0.60), (0.80, 3.20), (3.00, 15.00) USD per million tokens:

$$C(q, m) = \frac{t_q^{\text{in}} p_m^{\text{in}} + t_q^{\text{out}} p_m^{\text{out}}}{10^6} \quad (1)$$

$$\Delta C_\pi = 100 \frac{C_{\text{ref}} - \mathbb{E}_q[C(q, \pi(q))]}{C_{\text{ref}}}, \quad C_{\text{ref}} = \mathbb{E}_q[C(q, \text{frontier})] \quad (2)$$

$$Q_\pi = \mathbb{E}_q[\text{qual}(q, \pi(q))] \quad (3)$$

$$F(\pi) = 100 Q_\pi + 0.6 \Delta C_\pi - 0.05 L_\pi - 300 \mathbf{1}[L_\pi > 100\ \mu\text{s}] \quad (4)$$

where L_π is the mean wall-clock decision time in microseconds. Every candidate must pass three gates: **Gate 1** parses the source with `ast` and rejects imports, `exec/global` statements, dangerous identifiers (`open`, `eval`, `exec`, `os`, `sys`, `__import__`), dunder attribute access, and oversized programs (>300 AST nodes); survivors execute in a namespace whose builtins are a fixed safe subset. **Gate 2** runs eight boundary smoke tests (zero-token, 200k-token, math-only, code-only, extreme readability/density) requiring tier-valued string outputs. **Gate 3** is the vectorized benchmark scoring Q , ΔC , and L over the full training split with the hard latency penalty above. The corpus comprises 10,000 synthetic queries (30.1% code-syntax, 27.6% math-symbol, mean 184 tokens) with tier qualities drawn from difficulty-conditional distributions; splits are 60/40 train/test with a top-decile OOD slice (832 queries: extreme length, code/math, or density).

3 Evolutionary Search and MAP-Elites Topology

Search maintains three islands of 14 policies. Each generation, every island produces 8 offspring by tournament selection over its fitness-ranked members (15% of births are fresh grammar-template immigrants), applies one semantic mutation operator – numeric-boundary jitter, comparison flips, branch rotation, condition inversion, grammar synthesis, or two-parent branch crossover – each wrapped in a simulated $\langle \text{thought} \rangle / \langle \text{code} \rangle$ response with a deterministic fallback generator, and is pruned to the elite top- k . The fallback generator is a deliberate design choice, not a limitation: it guarantees compute-free, bit-reproducible exploration with zero calls to expensive closed LLM APIs, so that every result in this paper regenerates identically on a laptop; because mutants already flow through the $\langle \text{thought} \rangle / \langle \text{code} \rangle$ interface, swapping the fallback for LLM-in-the-loop mutation [7, 4] is a drop-in change we leave to future work. Every second generation the island best migrates cyclically. In parallel, a 6×6 MAP-Elites grid [5] bins candidates by quality $\in [0.40, 0.95]$ and cost reduction $\in [0, 100]\%$, retaining the fittest policy per cell as a quality-diversity archive; all evaluations, parent links, thoughts, and gate outcomes are persisted to SQLite (`evolution_search.db`), giving a fully replayable lineage. The reported Pareto frontier is the non-dominated set over the archive under $(\max Q, \max \Delta C)$ [6]. The reference run executes $10 \text{ generations} \times 3 \text{ islands} \times 8 \text{ offspring} = 240$ evaluations in roughly five seconds on a laptop CPU.

4 Empirical Evaluation and Baseline Comparison

Table 1 compares the evolved champion against constant, random, hand-engineered, and supervised (scikit-learn decision tree, `DecisionTreeClassifier(max_depth=8, min_samples_leaf=20)`, train accuracy 0.782) baselines. The supervised router collapses to near always-frontier behavior ($Q = 0.7203, \Delta C = 0.47\%$): imitating the quality-optimal label does not encode cost pressure, and its per-query scikit-learn inference costs $67.6 \mu\text{s} - 489\times$ the champion’s decision time for a strictly dominated operating point. The evolved champion instead retains 82.0% of frontier quality at 18.6% of frontier cost with a single effective decision threshold pair. Figure 1 visualizes the Table 1 comparison; Figure 2 shows the full quality-cost Pareto frontier of the archive.

Out-of-distribution robustness. On the OOD slice the champion’s cost reduction *rises* to 85.46% (from 81.36%) because hard long-tail queries escalate to medium rather than frontier, while the hand-written baseline sheds 39.6 points of cost efficiency ($72.35\% \rightarrow 32.77\%$). Quality gaps (ΔQ) are positive for the top-50 evolved policies, indicating the search did not overfit the train split’s difficulty mix.

Ablations. Table 2 isolates each component under an identical 24-evaluations-per-generation budget, with every variant replicated over five seeds. The headline finding is a negative result we consider equally informative: mean peak fitness is statistically indistinguishable across all variants (all means within 0.17 fitness of each other, $\sigma \leq 0.43$) – at this budget the framework, not the configuration, does the heavy lifting. Within that envelope, three seed-robust signals emerge. (C) *Operators* carry the only convergence effect: semantic mutation reaches 95% of final fitness in 1.0 ± 0.0 generations versus 1.6 ± 1.3 for random constant jitter, dominates the mean elite curve at every generation of the early search (+1.40 fitness at generation 1), and yields a tighter, higher-quality archive (223 ± 8 elites versus 244 ± 0 – jitter-only offspring are all novel but none is better), confirming that structured, thought-guided edits – not blind threshold noise – drive efficient search. (A) *Topology* leaves peak fitness unchanged (single-population delta -0.03, within 1σ) but

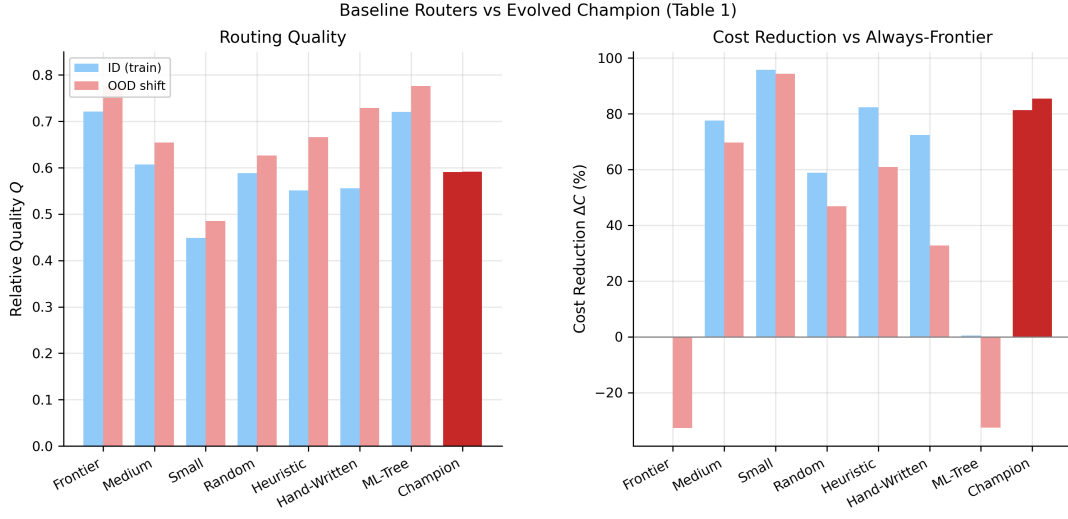


Figure 1: Baseline comparison across in-distribution (ID) and out-of-distribution (OOD) query slices. Left: relative quality Q ; right: cost reduction ΔC versus always-frontier routing. The evolved champion (red) occupies a Pareto-dominant efficiency operating point.

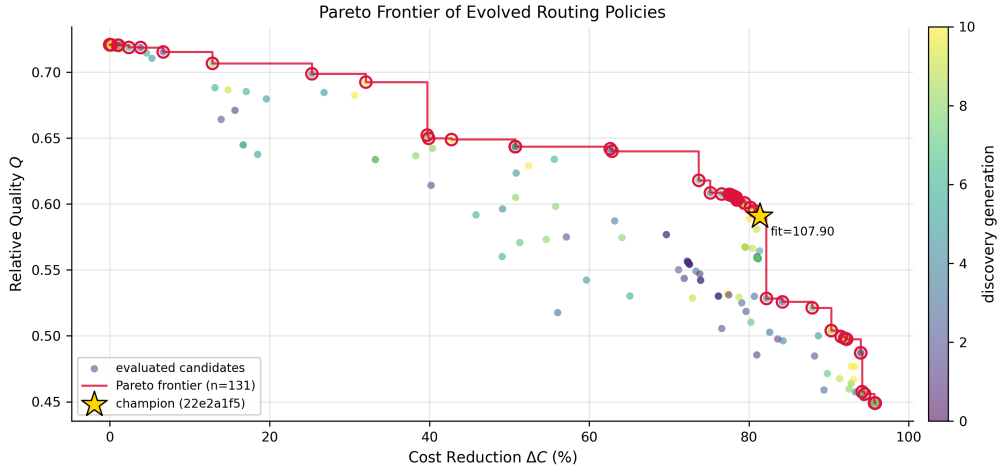


Figure 2: Pareto frontier of the evolved policy archive over relative quality Q and cost reduction ΔC ; points are colored by discovery generation and the star marks the released champion policy.

Table 1: Main results (Table 1). ID: in-distribution train split ($n=6000$); OOD: distribution-shifted slice ($n=832$). Reference cost: always-frontier.

Method	Q (ID)	$\Delta C\%$ (ID)	Q (OOD)	$\Delta C\%$ (OOD)	L (μs)	Params	Mem (KB)
Frontier	0.7208	0.00	0.7763	-32.74	0.100	1	0.23
Medium	0.6072	77.56	0.6544	69.69	0.097	1	0.23
Small	0.4490	95.79	0.4851	94.32	0.098	1	0.23
Random	0.5888	58.82	0.6266	46.79	0.419	0	0.32
Heuristic	0.5512	82.32	0.6661	60.85	0.250	6	0.81
Hand-Written	0.5560	72.35	0.7292	32.77	0.174	5	0.85
ML-Tree	0.7203	0.47	0.7761	-32.56	67.624	133	12.65
Champion	0.5909	81.36	0.5912	85.46	0.138	1	0.43

Table 2: Ablation results (Table 2), aggregated as mean $\pm$ std over 5 seeds (42, 123, 2026, 777, 999). Budget-controlled: every variant evaluates 24 offspring per generation for 10 generations. Gen $\rightarrow$ 95%: first generation reaching 95% of the run-final best fitness; Elites: unique passing candidates in the run archive; Cells: occupied MAP-Elites grid cells.

Axis	Variant	Peak Fitness	Gen $\rightarrow$ 95%	Gen $>$ Base	Elites	Cells
A (topology)	Single population	107.87 $\pm$ 0.36	1.0 $\pm$ 0.00	1.0 $\pm$ 0.00	220.0 $\pm$ 7.25	12.6 $\pm$ 0.89
A (topology)	3-island + migration (full)	107.85 $\pm$ 0.30	1.0 $\pm$ 0.00	1.0 $\pm$ 0.00	223.2 $\pm$ 7.73	12.8 $\pm$ 0.45
B (archive)	Greedy fitness replacement	107.89 $\pm$ 0.26	1.0 $\pm$ 0.00	1.0 $\pm$ 0.00	222.6 $\pm$ 6.66	0.0 $\pm$ 0.00
B (archive)	2D MAP-Elites parents	107.79 $\pm$ 0.34	1.0 $\pm$ 0.00	1.0 $\pm$ 0.00	222.6 $\pm$ 5.32	12.2 $\pm$ 0.84
C (operators)	Random constant jitter	107.72 $\pm$ 0.43	1.6 $\pm$ 1.34	1.0 $\pm$ 0.00	244.0 $\pm$ 0.00	10.6 $\pm$ 0.55
C (operators)	Semantic mutation (full)	107.85 $\pm$ 0.30	1.0 $\pm$ 0.00	1.0 $\pm$ 0.00	223.2 $\pm$ 7.73	12.8 $\pm$ 0.45

cuts frontier-size variance by 65% (σ 28.2 $\rightarrow$ 9.9 non-dominated policies), stabilizing deployable trade-off coverage across reruns. (B) The *archive* axis is fitness-equivalent (MAP-Elites delta -0.10, within 1σ); its value lies in the explicit quality-diversity grid itself – an inspectable, bounded archive of the trade-off surface – rather than in peak-fitness gains. We report these effect sizes with their seeds to guard against the single-seed over-interpretation common in evolutionary-computation studies. Figure 3 plots the corresponding per-island elite and mean fitness trajectories.

5 Mechanistic Deconstruction of Champion Policy 22e2a1f5

The champion (generation 10, island 2, parent 7f11eb26, operator `tweak_constants`, fitness 107.90) is:

```
def select_model(q):
    if not (not q["has_math_symbols"]):
        return "medium"
    if not (q["estimated_tokens"] > 367):
        return "medium"
    return "small"
```

The double negations look puzzling but are logically transparent: $\neg\neg p \equiv p$, so the first guard reads “if the query has math symbols” and the second reads “if the query has at most 367 estimated tokens.” Each redundant negation costs a nanosecond-scale boolean evaluation and nothing in fitness, which is precisely why selection never removed them – a neutral polymorphism fixed by drift. The compiled semantics are therefore: math-symbol queries $\rightarrow$ medium; non-math queries

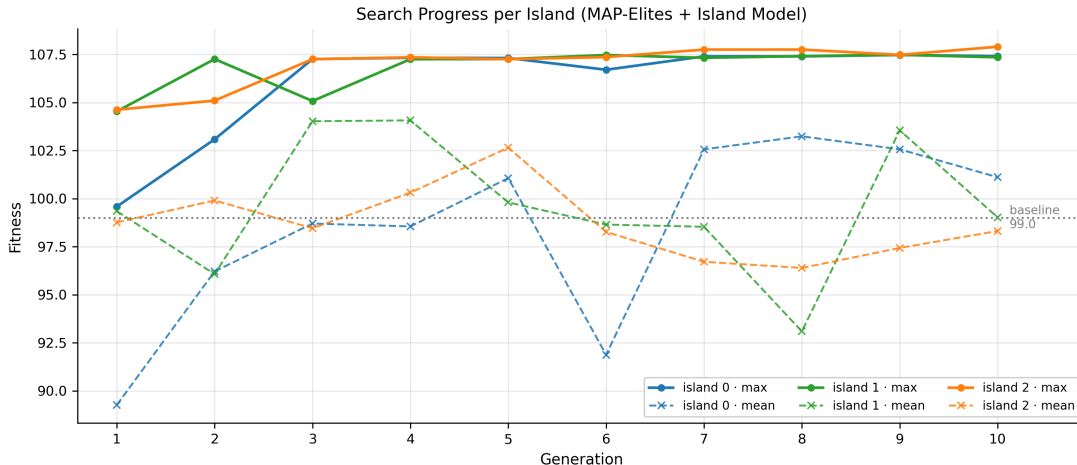


Figure 3: Search progress per island across ten generations: solid lines show the per-generation elite (best) fitness, dashed lines the mean fitness of newly registered candidates; the dotted line marks the hand-written baseline.

with at most 367 estimated tokens $\rightarrow$ medium; all other (long, non-math) queries $\rightarrow$ small; the frontier branch is unreachable. Three mechanisms explain its fitness: (1) *frontier elimination* – the $20\times$ price premium is never paid, yielding $\Delta C \approx 81\%$; (2) *difficulty-aware escalation* – math queries carry the largest frontier–small quality gap in the corpus, and medium captures most of it at a quarter of frontier price; (3) *difficulty inversion for long queries* – long non-math queries (chat/analysis) have below-average difficulty, so bulk traffic routes to the small tier at almost no quality cost. The lineage database shows the champion’s 367-token threshold emerged as a `tweak_constants` jitter of its parent’s 522-token threshold, one generation after branch rotation reordered the two guards – a textbook example of neutral drift followed by exploitative refinement.

6 Limitations, OOD Generalization, and Conclusion

Limitations. (i) The corpus and per-tier qualities are synthetic; absolute numbers will differ on live API traffic, though the *relative* baseline ordering and the mechanistic findings depend only on the ordinal difficulty structure. (ii) Latency L measures policy decision overhead, not end-to-end model latency. (iii) The five-seed replication varies search stochasticity only; the corpus itself is a single synthetic draw and should be regenerated across corpus seeds. (iv) The 6×6 grid is coarse; finer binning may expose additional niches. (v) The OOD shift is feature-level (extreme tokens, `code^math`, density), not adversarial.

Toward live production deployment. The bridge from synthetic benchmark to production proceeds in three concrete stages. **Stage 1 – Trace Replay on live logs:** re-score anonymized production queries through all three tiers to replace the simulated quality model with empirical per-tier quality, then re-run the seconds-cheap search on the calibrated corpus. **Stage 2 – Stratified Human/LLM Judgment:** within each difficulty stratum, compare champion versus always-frontier with human raters or an LLM judge, validating the ordinal quality structure the policy exploits. **Stage 3 – Shadow Canary Deployment:** mirror 1–5% of traffic to the evolved policy

while monitoring per-tier quality drift, cost per resolved query, and tail latency, with automatic rollback and scheduled re-evolution as model prices and capabilities drift.

Conclusion. Casting LLM routing as evolutionary code search over a deterministic multi-gate oracle yields an auditable, near-zero-latency policy that dominates hand-designed and supervised baselines: 81.4% cost reduction at $0.138\mu\text{s}$ per decision with positive OOD generalization. The full lineage, ablation harness, and paper-generation pipeline are deterministic and re-executable end-to-end. Future work: LLM-in-the-loop mutation replacing the fallback generator [7, 4], validation on live RouteLLM-style preference data [2], and multi-objective selection directly on the Pareto front [6].

Competing interests. The author declares no competing interests. Correspondence should be addressed to Thomas Gia Vy Day (giavytday@gmail.com).

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